

- Sedation, dizziness, sleep disturbance, headache, impaired concentration
- Dry mouth, nausea, vomiting, constipation, anorexia
- Impotence
- Tardive dyskinesia
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- Rare hypomania
- Palpitations, hypertension
- Weight gain



### **Life-Threatening or Dangerous Side Effects**

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

## **Weight Gain**



- Many experience and/or can be significant in amount

## **Sedation**



- Many experience and/or can be significant in amount, especially at high doses

## **What to Do About Side Effects**

- Wait
- Wait
- Wait
- For drug-induced parkinsonism, add an anticholinergic agent
- Beta blockers, benzodiazepines, or 5HT<sub>2A</sub> antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Reduce the dose
- For sedation, give at night
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

## **Best Augmenting Agents for Side Effects**

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT<sub>2A</sub> antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

## **Dosing and Use**

### **Usual Dosage Range**

- Schizophrenia: 400–800 mg/day in 2 doses (oral)
- Predominantly negative symptoms: 50–300 mg/day (oral)
- Intramuscular injection: 600–800 mg/day
- Depression: 150–300 mg/day (oral)

### **Dosage Forms**

- Different formulations may be available in different markets
- Tablet 200 mg, 400 mg, 500 mg
- Intramuscular injection 50 mg/mL, 100 mg/mL

### **How to Dose**

- Initial 400–800 mg/day in 1–2 doses; may need to increase dose to control positive symptoms; maximum generally 2400 mg/day

### **Dosing Tips**

- \* Low doses of sulpiride may be more effective at reducing negative symptoms than positive symptoms in schizophrenia; high doses may be equally effective at reducing both symptom dimensions
- \* Lower doses are more likely to be activating; higher doses are more likely to be sedating
- Some patients receive more than 2400 mg/day
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below  $1000/\text{mm}^3$

### **Overdose**

- Can be fatal; vomiting, agitation, hypotension, hallucinations, CNS depression, sinus tachycardia, arrhythmia, dystonia, dysarthria, hyperreflexia

### **Long-Term Use**

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

### **Habit Forming**

- No

### **How to Stop**

- Recommended to reduce dose over a week
- Slow down-titration (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after sulpiride is discontinued

### **Pharmacokinetics**

- Elimination half-life approximately 6–8 hours
- Excreted largely unchanged



### **Drug Interactions**

- Sulpiride may increase the effects of antihypertensive drugs

- CNS effects may be increased if sulpiride is used with other CNS depressants
- May decrease the effects of levodopa, dopamine agonists
- Antacids or sucralfate may reduce the absorption of sulpiride

### **Other Warnings/Precautions**

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold
- Antiemetic effect of sulpiride may mask signs of other disorders or overdose; suppression of cough reflex may cause asphyxia
- Use with caution in patients with hypertension, cardiovascular disease, pulmonary disease, hyperthyroidism, urinary retention, glaucoma
- May exacerbate symptoms of mania or hypomania
- Use only with caution if at all in Parkinson's disease or Lewy body dementia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy

### **Do Not Use**

- If patient has pheochromocytoma
- If patient has prolactin-dependent tumor
- If patient is pregnant or nursing
- In children under age 15
- If there is a proven allergy to sulpiride

## **Special Populations**

### **Renal Impairment**

- Use with caution; drug may accumulate
- Sulpiride is eliminated by the renal route; in cases of severe renal insufficiency, the dose should be decreased and intermittent treatment or switching to another antipsychotic should be considered

### **Hepatic Impairment**

- Use with caution

### **Cardiac Impairment**

- Use with caution

### **Elderly**

- Some patients may tolerate lower doses better
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events



### **Children and Adolescents**

- Not recommended for use in children under age 15
- 14 and older: recommended 3–5 mg/kg per day



### **Pregnancy**

- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Potential risks should be weighed against the potential benefits, and sulpiride should be used only if deemed necessary
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary

- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy

### **Breast Feeding**

- Some drug is found in mother's breast milk
- \* Recommended either to discontinue drug or bottle feed
- Immediate postpartum period is a high-risk time for relapse of psychosis

## **The Art of Psychopharmacology**

### **Potential Advantages**

- For negative symptoms in some patients

### **Potential Disadvantages**

- Patients who cannot tolerate sedation at high doses
- Patients with severe renal impairment

### **Primary Target Symptoms**

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive functioning

- Depressive symptoms
- Aggressive symptoms

### **Pearls**

- \* There is some controversy over whether sulpiride is more effective than older conventionals at treating negative symptoms
- Sulpiride has been used to treat migraine associated with hormonal changes
- \* Some patients with inadequate response to clozapine may benefit from augmentation with sulpiride
- Sulpiride is poorly absorbed from the gastrointestinal tract and penetrates the blood–brain barrier poorly, which can lead to highly variable clinical responses, especially at lower doses
- Small studies and clinical anecdotes suggest efficacy in depression and anxiety disorders (“neuroses”) at low doses
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical antipsychotics where antipsychotic responses of individual patients can occasionally vary greatly from one atypical antipsychotic to another
- Patients with inadequate responses to atypical antipsychotics may benefit from a trial of augmentation with a conventional antipsychotic such as sulpiride or from switching to a conventional antipsychotic such as sulpiride

- However, long-term polypharmacy with a combination of a conventional antipsychotic with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

### *Suggested Reading*

Caley CF , Weber SS . Sulpiride: an antipsychotic with selective dopaminergic antagonist properties . *Ann Pharmacother* 1995 ;**29** (2 ):152–60 .

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Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Omori IM , Wang J. Sulpiride versus placebo for schizophrenia . *Cochrane Database Syst Rev* 2009 ;**15** (2 ):CD007811 .  
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Wang J , Omori IM , Fenton M , Soares B. Sulpiride augmentation for schizophrenia . *Cochrane Database Syst Rev* 2010 ;**20** (1 ):CD008125 .

Suvorexant

## Therapeutics

### Brands

- Belsomra

### Generic?

No

### Class

- Dual orexin receptor antagonist (DORA); hypnotic

### Commonly Prescribed for

*(bold for FDA approved)*

- **Insomnia (problems with sleep onset and/or sleep maintenance)**

### How the Drug Works

- Orexin serves to stabilize and promote wakefulness; suvorexant binds to orexin 1 and orexin 2 receptors, blocking orexin from binding there and thus preventing it from promoting wakefulness

### How Long Until It Works

- Generally takes effect in less than an hour

### **If It Works**

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings could theoretically be decreased over time

### **If It Doesn't Work**

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene
- Switch to another agent



### **Best Augmenting Combos for Partial Response or Treatment Resistance**

- No controlled trials of combinations with other hypnotics or psychotropic drugs
- Generally, best to switch to another agent

### **Tests**

- None for healthy individuals

# Side effects

## How Drug Causes Side Effects

- Theoretically due to downstream effects of blocking orexin receptors

## Notable Side Effects

- Sedation, headache, dizziness, abnormal dreams

## Life-Threatening or Dangerous Side Effects

- Sleep paralysis and hypnagogic/hypnopompic hallucinations (rare)
- Dose-dependent symptoms similar to mild cataplexy (rare)



## Weight Gain

- Reported but not expected



## Sedation

- Many experience and/or can be significant in amount

## What to Do About Side Effects

- Wait
- To avoid problems with memory, take suvorexant only if planning to have a full night's sleep
- Lower the dose

- Switch to a different hypnotic

### **Best Augmenting Agents for Side Effects**

- Many side effects cannot be improved with an augmenting agent

## **Dosing and use**

### **Usual Dosage Range**

- 10 mg/night

### **Dosage Forms**

- Tablet 5 mg, 10 mg, 15 mg, 20 mg

### **How to Dose**

- Starting dose is 10 mg, no more than once per night and within 30 minutes of bedtime
- Should not take unless there are at least 7 hours remaining of sleep time



### **Dosing Tips**

- Patients who tolerate but do not respond to 10 mg may receive 15 mg or 20 mg doses; 20 mg is the maximum recommended dose
- Use the lowest dose effective for the patient

- Taking suvorexant with or soon after a meal can delay the time to effect
- Not restricted to short-term use

### **Overdose**

- Limited data; sedation

### **Long-Term Use**

- Has been evaluated and found effective in trials up to 1 year

### **Habit Forming**

- Suvorexant is a Schedule IV drug
- There was no evidence of physiological dependence or withdrawal symptoms with prolonged use of suvorexant

### **How to Stop**

- Taper not necessary

### **Pharmacokinetics**

- Metabolized by CYP450 3A4
- Mean terminal half-life approximately 12 hours

### **Drug Interactions**

- Not recommended in patients taking concomitant strong CYP450 3A4 inhibitors
- Patients taking moderate CYP3A4 inhibitors should receive a 5 mg dose; dose can be increased to 10 mg if needed
- Patients taking CYP3A4 inducers may experience reduced efficacy of suvorexant
- Monitor digoxin levels if coadministered with suvorexant
- Increased depressive effects when taken with other CNS depressants

### **Other Warnings/Precautions**

- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with caution in patients with impaired respiratory function or obstructive sleep apnea
- Suvorexant should only be administered at bedtime
- Label contains warning for risk of next-day impaired alertness and motor coordination

- Sleep driving and other complex behaviors, such as eating and preparing food and making phone calls, have been reported in patients taking sedative hypnotics; if complex sleep behaviors occur, discontinue use

### **Do Not Use**

- If patient has narcolepsy
- If patient is taking strong CYP450 3A4 inhibitors
- If there is a proven allergy to suvorexant

## **Special populations**

### **Renal Impairment**

- Dose adjustment not necessary

### **Hepatic Impairment**

- Dose adjustment not necessary in mild to moderate hepatic impairment
- Not recommended in patients with severe hepatic impairment

### **Cardiac Impairment**

- Not studied in patients with cardiac impairment

### **Elderly**

- Some patients may tolerate lower doses better



### **Children and Adolescents**

- Safety and efficacy have not been established



### **Pregnancy**

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- When administered during organogenesis, the no observed adverse effect levels (NOAELs) for fetal toxicity are 25 and 28 times the maximum recommended human dose (MRHD) based on AUC in rats and rabbits, respectively
- When administered during pregnancy and lactation, the NOAEL for developmental toxicity is 25 times the MRHD based on AUC in rats

### **Breast Feeding**

- Unknown if suvorexant is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Recommended to either discontinue drug or bottle feed

# **The art of psychopharmacology**

## **Potential Advantages**

- Primary insomnia
- Chronic insomnia
- Those who require long-term treatment
- Those with depression whose insomnia does not resolve with antidepressant treatment

## **Potential Disadvantages**

- More expensive than some other sedative hypnotics

## **Primary Target Symptoms**

- Time to sleep onset
- Nighttime awakenings
- Total sleep time



## **Pearls**

- DORAs are reversible inhibitors of orexin, so their action can theoretically be reversed as endogenous orexin is released upon awakening, and even to a certain extent in the middle of the night
- This is unlike Z-drug and benzodiazepine hypnotics, which do not reverse upon wakening

- DORAs hypothetically switch off alertness-related circuits, countering inappropriate wakefulness associated with insomnia, whereas Z-drug and benzodiazepine hypnotics and melatonin hypothetically switch on sleep-related circuits
- Curbing alertness-related factors in patients with insomnia may have some theoretical efficacy and tolerability advantages over enhancing sleep drives, sedation, and somnolence
- Targeting insomnia may prevent the onset of major depressive disorder (MDD) or generalized anxiety disorder (GAD) and help maintain remission after recovery from MDD or GAD
- Rebound insomnia does not appear to be common
- Suvorexant may be safe for long-term use, with little or no suggestion of tolerance, dependence, or abuse
- May even be safe to consider in patients with a past history of substance abuse who require treatment with a hypnotic
- May be effective in patients with insomnia unresponsive to medications with other mechanisms of action (e.g., Z-drug hypnotics, benzodiazepines)

### *Suggested Reading*

Citrome L. Suvorexant for insomnia: a systematic review of the efficacy and safety profile for this newly approved hypnotic –what is the number

needed to treat, number needed to harm and likelihood to be helped or harmed? *Int J Clin Pract* 2014 ;**68** (12 ):1429 –41.

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Michelson D , Snyder E , Paradis E , et al. Safety and efficacy of suvorexant during 1-year treatment of insomnia with subsequent abrupt treatment discontinuation: a phase 3 randomised, double-blind, placebo-controlled trial . *Lancet Neurol* 2014 ;**13** (5 ):461–71 .

Tasimelteon

## Therapeutics

### Brands

- Hetlioz

### Generic?

No

### Class

- Melatonin 1 and 2 receptor agonist

### Commonly Prescribed for

*(bold for FDA approved)*

- **Non-24-Hour Sleep-Wake Disorder (Non-24)**
- Insomnia associated with shift work, jet lag, or circadian rhythm disturbances

### How the Drug Works

- Binds selectively to melatonin 1 and melatonin 2 receptors as a full agonist, with greater affinity for melatonin 2 receptors than for melatonin 1 receptors

### How Long Until It Works

- Because of individual differences in circadian rhythms, daily use for several weeks or months may be necessary before efficacy is

observed

### **If It Works**

- Causes circadian entrainment
- Increases nighttime sleep time
- Reduces daytime nap time

### **If It Doesn't Work**

- Switch to another agent

### **Best Augmenting Combos for Partial Response or Treatment Resistance**

- No controlled trials of combinations with other hypnotics or psychotropic drugs
- Generally, best to switch to another agent

### **Tests**

- None for healthy individuals

## **Side effects**

### **How Drug Causes Side Effects**

- Actions at melatonin receptors that carry over to the next day could theoretically cause daytime sedation, fatigue, and sluggishness, but this is not common